

Chapter 4

Saturday, November 09, 2024

12:40 PM

Topics:

- Gaussian Elimination

Gaussian Elimination

Pivot equation Pivot coefficient

$$\begin{aligned} & 4x_1 - 2x_2 - 3x_3 + 6x_4 = 12 \\ & 6x_1 + 7x_2 + 6.5x_3 - 6x_4 = -6.5 \\ & 1x_1 + 7.5x_2 + 6.25x_3 + 5.5x_4 = 16 \\ & -12x_1 + 22x_2 + 15.5x_3 - 1x_4 = 17 \end{aligned}$$

$m_{21} = \frac{a_{21}}{a_{11}} = \frac{-6}{4} = -1.5$

$$\begin{aligned} & -1.5 \times [4x_1 - 2x_2 - 3x_3 + 6x_4 = 12] \\ & = -6x_1 + 3x_2 + 4.5x_3 - 9x_4 = -18 \end{aligned}$$
$$\begin{aligned} & [-6x_1 + 7x_2 + 6.5x_3 - 6x_4 = -6.5] \\ & [-6x_1 + 3x_2 + 4.5x_3 - 9x_4 = -18] \\ & \quad 0 + 4x_2 + 2x_3 + 3x_4 = 11.5 \end{aligned}$$

$m_{31} = \frac{a_{31}}{a_{11}} = \frac{1}{4} = 0.25$

$$\begin{aligned} & 0.25 \times [4x_1 - 2x_2 - 3x_3 + 6x_4 = 12] \\ & = 1x_1 - 0.5x_2 - 0.75x_3 + 1.5x_4 = 3 \end{aligned}$$
$$\begin{aligned} & [1x_1 + 7.5x_2 + 6.25x_3 + 5.5x_4 = 16] \\ & - [1x_1 - 0.5x_2 - 0.75x_3 + 1.5x_4 = 3] \\ & \quad 0 + 8x_2 + 7x_3 + 4x_4 = 13 \end{aligned}$$

$m_{41} = \frac{a_{41}}{a_{11}} = \frac{-12}{4} = -3$

$$\begin{aligned} & -3 \times [4x_1 - 2x_2 - 3x_3 + 6x_4 = 12] \\ & = -12x_1 + 6x_2 + 9x_3 - 18x_4 = -36 \end{aligned}$$
$$\begin{aligned} & [-12x_1 + 22x_2 + 15.5x_3 - 1x_4 = 17] \\ & - [-12x_1 + 6x_2 + 9x_3 - 18x_4 = -36] \\ & \quad 0 + 16x_2 + 6.5x_3 + 17x_4 = 53 \end{aligned}$$

Gaussian Elimination

$$\begin{aligned}
 4x_1 - 2x_2 - 3x_3 + 6x_4 &= 12 \\
 + 4x_2 + 2x_3 + 3x_4 &= 11.5 \quad \text{Pivot equation} \quad \text{Pivot coefficient} \\
 + 8x_2 + 7x_3 + 4x_4 &= 13 \\
 + 16x_2 + 6.5x_3 + 17x_4 &= 53
 \end{aligned}$$

$$m_{32} = \frac{a_{32}}{a_{22}} = \frac{8}{4} = 2$$

$$2 \times [4x_2 + 2x_3 + 3x_4 = 11.5] = 8x_2 + 4x_3 + 6x_4 = 23$$

$$[8x_2 + 7x_3 + 4x_4 = 13] - [8x_2 + 4x_3 + 6x_4 = 23] = 0 + 3x_3 - 2x_4 = -10$$

$$m_{42} = \frac{a_{42}}{a_{22}} = \frac{16}{4} = 4$$

$$4 \times [4x_2 + 2x_3 + 3x_4 = 11.5] = 16x_2 + 8x_3 + 12x_4 = 46$$

$$[16x_2 + 6.5x_3 + 17x_4 = 53] - [16x_2 + 8x_3 + 12x_4 = 46] = 0 - 1.5x_3 + 5x_4 = 7$$

$$\begin{aligned}
 4x_1 - 2x_2 - 3x_3 + 6x_4 &= 12 \\
 + 4x_2 + 2x_3 + 3x_4 &= 11.5 \\
 + 3x_3 - 2x_4 &= -10 \quad \text{Pivot equation} \quad \text{Pivot coefficient} \\
 -1.5x_3 + 5x_4 &= 7
 \end{aligned}$$

$$m_{43} = \frac{a_{43}}{a_{33}} = \frac{-1.5}{3} = -0.5$$

$$-0.5 \times [3x_3 - 2x_4 = -10] = -1.5x_3 + 1x_4 = 5$$

$$[-1.5x_3 + 5x_4 = 7] - [-1.5x_3 + 1x_4 = 5] = 0 + 4x_4 = 2$$

Upper Triangular Matrix

$$\begin{aligned}
 4x_1 - 2x_2 - 3x_3 + 6x_4 &= 12 \\
 + 4x_2 + 2x_3 + 3x_4 &= 11.5 \\
 + 3x_3 - 2x_4 &= -10 \\
 + 4x_4 &= 2
 \end{aligned}$$

$$\begin{aligned}
 x_4 &= 0.5 \\
 3x_3 - 1 &= -10 \rightarrow x_3 = -3 \\
 4x_2 - 6 + 1.5 &= 11.5 \rightarrow x_2 = 4 \\
 4x_1 - 8 + 9 + 3 &= 12 \rightarrow x_1 = 2
 \end{aligned}$$

- Gauss-Jordan Method

Gauss-Jordan Elimination

$$\begin{aligned}
 4x_1 - 2x_2 - 3x_3 + 6x_4 &= 12 \\
 + 4x_2 + 2x_3 + 3x_4 &= 11.5 \\
 + 8x_2 + 7x_3 + 4x_4 &= 13 \\
 + 16x_2 + 6.5x_3 + 17x_4 &= 53
 \end{aligned}$$

Pivot element	4	-2	-3	6	12		Pivot row
	-6	7	6.5	-6	-6.5		
	1	7.5	6.25	5.5	16		
	-12	22	15.5	-1	17		

Pivot element	1	7.5	6.25	5.5	16		Pivot row
	-6	7	6.5	-6	-6.5		
	1	7.5	6.25	5.5	16		
	-12	22	15.5	-1	17		

1	-0.5	-0.75	1.5	3		Pivot row
-6	7	6.5	-6	-6.5	$\leftarrow (-6) \times [1 -0.5 -0.75 1.5 3]$	
1	7.5	6.25	5.5	16	$\leftarrow (1) \times [1 -0.5 -0.75 1.5 3]$	
-12	22	15.5	-1	17	$\leftarrow (-12) \times [1 -0.5 -0.75 1.5 3]$	

Pivot element	1	-0.5	-0.75	1.5	3		Pivot row
	0	14	2	3	11.5		
	0	8	7	4	13		
	0	16	6.5	17	53		

1	-0.5	-0.75	1.5	3		Pivot row
0	14	2	3	11.5		
0	8	7	4	13		
0	16	6.5	17	53		

Craft's Method (no Gaussian Elimination $\rightarrow$ Matrix multiplication)

$$[A] = [L][U]$$

$$[A] = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 20 & 26 \\ 3 & 26 & 70 \end{bmatrix} = [L][U] \quad [L] = \begin{bmatrix} L_{11} & 0 & 0 \\ L_{21} & L_{22} & 0 \\ L_{31} & L_{32} & L_{33} \end{bmatrix} \quad [U] = \begin{bmatrix} 1 & U_{12} & U_{13} \\ 0 & 1 & U_{23} \\ 0 & 0 & 1 \end{bmatrix}$$

$a_{11} \quad 1 = [L_{11} \ 0 \ 0] \times \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \quad L_{11} + 0 + 0 \therefore L_{11} = 1$
 $a_{12} \quad 2 = [1 \ 0 \ 0] \times \begin{bmatrix} 0 \\ U_{12} \\ 0 \end{bmatrix} \quad (U_{12} \times 1) + (0 \times 1) + (0 \times 0) \therefore U_{12} = 2$
 $a_{13} \quad 3 = [1 \ 0 \ 0] \times \begin{bmatrix} 0 \\ 0 \\ U_{13} \end{bmatrix} \quad (U_{13} \times 1) + (0 \times 2) + (0 \times 0) \therefore U_{13} = 3$
 $a_{21} \quad 2 = [L_{21} \ L_{22} \ 0] \times \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \quad (L_{21} \times 1) + (L_{22} \times 0) + (0 \times 0) \therefore L_{21} = 2$
 $a_{22} \quad 20 = [2 \ L_{22} \ 0] \times \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \quad (2 \times 2) + (L_{22} \times 1) + (0 \times 0) \therefore L_{22} = 16$
 ~~$a_{23} \quad 26 = [2 \ 16 \ 0] \times \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \quad (2 \times 3) + (16 \times 3) + (0 \times 1) \therefore 6 + 48 = 54 \neq 26$~~
 $a_{23} \quad 26 = [2 \ 16 \ 0] \times \begin{bmatrix} 0 \\ U_{23} \\ 1 \end{bmatrix} \quad 6 + U_{23} \cdot 16 \therefore U_{23} = \frac{20}{16} = \frac{5}{4}$
 $a_{31} \quad 3 = [L_{31} \ L_{32} \ L_{33}] \times \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \therefore L_{31} = 3$
 $a_{32} \quad 26 = [3 \ L_{32} \ L_{33}] \times \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \quad 6 + L_{32} \therefore L_{32} = 20$
 $a_{33} \quad 70 = [3 \ 20 \ L_{33}] \times \begin{bmatrix} 0 \\ 5/4 \\ 1 \end{bmatrix} \quad 9 + 25 + L_{33} \therefore L_{33} = 36$

Craft's Method

$$[L] = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 16 & 0 \\ 3 & 20 & 36 \end{bmatrix} \quad [U] = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 5/4 \\ 0 & 0 & 1 \end{bmatrix}$$

- Matrix inversion by LU decomposition

$$\left[\begin{array}{ccc|ccc} 10 & 12 & 0 & 1 & 0 & 0 \\ 0 & 2 & 8 & 0 & 1 & 0 \\ 2 & 4 & 8 & 0 & 0 & 1 \end{array} \right] \xrightarrow{R_1 = R_1 \div 10} \left[\begin{array}{ccc|ccc} 1 & 1.2 & 0 & 0.1 & 0 & 0 \\ 0 & 2 & 8 & 0 & 1 & 0 \\ 2 & 4 & 8 & 0 & 0 & 1 \end{array} \right] \xrightarrow{R_3 = R_3 - 2R_1} \left[\begin{array}{ccc|ccc} 1 & 1.2 & 0 & 0.1 & 0 & 0 \\ 0 & 2 & 8 & 0 & 1 & 0 \\ 0 & 1.6 & 8 & -0.2 & 0 & 1 \end{array} \right]$$

$$\xrightarrow{R_2 = R_2 \div 2} \left[\begin{array}{ccc|ccc} 1 & 1.2 & 0 & 0.1 & 0 & 0 \\ 0 & 1 & 4 & 0 & 0.5 & 0 \\ 0 & 1.6 & 8 & -0.2 & 0 & 1 \end{array} \right] \xrightarrow{R_1 = R_1 - (1.2 \times R_2)} \left[\begin{array}{ccc|ccc} 1 & 0 & -4.8 & 0.1 & -0.6 & 0 \\ 0 & 1 & 4 & 0 & 0.5 & 0 \\ 0 & 1.6 & 8 & -0.2 & 0 & 1 \end{array} \right] \xrightarrow{R_3 = R_3 - (1.6 \times R_2)} \left[\begin{array}{ccc|ccc} 1 & 0 & -4.8 & 0.1 & -0.6 & 0 \\ 0 & 1 & 4 & 0 & 0.5 & 0 \\ 0 & 0 & 1.6 & -0.2 & -0.8 & 1 \end{array} \right] \xrightarrow{R_3 = R_3 \times \frac{1}{1.6}}$$

$$\left[\begin{array}{ccc|ccc} 1 & 0 & -4.8 & 0.1 & -0.6 & 0 \\ 0 & 1 & 4 & 0 & 0.5 & 0 \\ 0 & 0 & 1 & -\frac{1}{8} & -\frac{1}{2} & \frac{5}{8} \end{array} \right] \xrightarrow{R_2 = R_2 - 4R_3} \left[\begin{array}{ccc|ccc} 1 & 0 & -4.8 & \frac{1}{10} & -\frac{6}{10} & 0 \\ 0 & 1 & 0 & \frac{1}{2} & \frac{5}{2} & \frac{1}{2} \\ 0 & 0 & 1 & -\frac{1}{8} & -\frac{1}{2} & \frac{5}{8} \end{array} \right]$$

$$\xrightarrow{R_1 = R_1 + 4.8R_3} \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & -\frac{1}{2} & -3 & 3 \\ 0 & 1 & 0 & \frac{1}{2} & \frac{5}{2} & \frac{5}{2} \\ 0 & 0 & 1 & -\frac{1}{8} & -\frac{1}{2} & \frac{5}{8} \end{array} \right]$$

4.13 Gauss
Jordan
Inversion

- Jacobi Method

Jacobi Method (Diagonally Dominant)

$$\begin{bmatrix} 9 & 2 & 3 \\ 1 & 12 & 9 \\ 4 & 6 & 14 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 7 \\ 2 \\ 1 \end{bmatrix} \quad \text{Initial guess} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

Iteration 1:

$$x_1 = (7 - [2x_2 + 3x_3]) / 9 = \frac{2}{9}$$

$$x_2 = (2 - [1x_1 + 9x_3]) / 12 = -\frac{2}{3}$$

$$x_3 = (1 - [4x_1 + 6x_2]) / 14 = \frac{9}{14}$$

Iteration 2:

$$x_1 = (7 - [2 \cdot \frac{2}{3} + 3 \cdot \frac{9}{14}]) / 9 = 1.14$$

$$x_2 = (2 - [1 \cdot \frac{2}{9} + 9 \cdot \frac{9}{14}]) / 12 = 0.63$$

$$x_3 = (1 - [4 \cdot \frac{2}{9} + 6 \cdot \frac{2}{3}]) / 14 = 0.29$$

Iteration 3:

$$x_1 = (7 - [2 \cdot 0.63 + 3 \cdot 0.29]) / 9 = 0.541$$

$$x_2 = (2 - [1 \cdot 1.14 + 9 \cdot 0.29]) / 12 = -0.1458$$

$$x_3 = (1 - [4 \cdot 1.14 + 6 \cdot 0.63]) / 14 = -0.524$$

Iteration 4:

$$x_1 = (7 - [2 \cdot -0.1458 + 3 \cdot -0.524]) / 9 = 0.9848$$

$$x_2 = (2 - [1 \cdot 0.541 + 9 \cdot -0.524]) / 12 = 0.51458$$

$$x_3 = (1 - [4 \cdot 0.541 + 6 \cdot -0.1458]) / 14 = -0.020657$$

Iteration 5:

$$x_1 = (7 - [2 \cdot 0.51458 + 3 \cdot -0.020657]) / 9 = 0.67$$

$$x_2 = (2 - [1 \cdot 0.9848 + 9 \cdot -0.020657]) / 12 = 0.1$$

$$x_3 = (1 - [4 \cdot 0.9848 + 6 \cdot 0.51458]) / 14 = -0.43$$

- Gauss-Seidel method

Gauss-Seidel Method faster convergence

Iteration 1:

$$x_1 = (7 - (2 \times 1 + 3 \times 1)) / 9 = \frac{2}{9}$$

$$x_2 = (2 - (1 \times \frac{2}{9} + 9 \times 1)) / 12 = -0.6$$

$$x_3 = (1 - (4 \times \frac{2}{9} + 6 \times -0.6)) / 14 = 0.265$$

Update equations with new answers

Iteration 2:

$$x_1 = (7 - (2 \times -0.6 + 3 \times 0.265)) / 9 = 0.8227$$

$$x_2 = (2 - (1 \times 0.8227 + 9 \times 0.265)) / 12 = -0.1006$$

$$x_3 = (1 - (4 \times 0.8227 + 6 \times -0.1006)) / 14 = -0.1205$$

Iteration 3:

- Thomas Algorithm

Thomas Algorithm

$$\begin{aligned} x_1 + x_2 &= 6 \\ 2x_1 + 7x_2 + 8x_3 &= 9 \\ 3x_2 + 5x_3 &= 6 \end{aligned} \quad \begin{bmatrix} 1 & 1 & 0 \\ 2 & 7 & 8 \\ 0 & 3 & 5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 6 \\ 9 \\ 6 \end{bmatrix}$$

zeros in upper and lower diagonal

$$\begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 0 & 3 & 1 \end{bmatrix} \begin{bmatrix} u_1 & 1 & 0 \\ 0 & u_2 & 8 \\ 0 & 0 & u_3 \end{bmatrix} = \begin{bmatrix} a_1 & b_1 & 0 \\ c_2 & a_2 & b_2 \\ 0 & c_3 & a_3 \end{bmatrix}$$

for $i=1, 2, \dots, n$ $l_i = \frac{c_i}{u_i - 1}$, $u_i = a_i - (l_i b_i - 1)$

$i=1$ $u_1 = a_1 = 1$

$i=2$ $l_2 = \frac{2}{1} = 2$, $u_2 = 7 - (2 \times 1) = 5$

$i=3$ $l_3 = \frac{3}{5} = 0.6$, $u_3 = 5 - (0.6 \times 8) = 0.2$

$$L = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 0 & 0.6 & 1 \end{bmatrix} \times U = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 5 & 8 \\ 0 & 0 & 0.2 \end{bmatrix} \quad \begin{bmatrix} 6 \\ 9 \\ 6 \end{bmatrix} \quad \begin{matrix} l_y = b \\ Ux = y \end{matrix}$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 0 & 3 & 1 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = \begin{bmatrix} 6 \\ 9 \\ 6 \end{bmatrix} \quad \begin{matrix} y_1 = 6 \\ y_2 = -3 \\ y_3 = \frac{39}{5} \end{matrix}$$

$$\begin{bmatrix} 1 & 1 & 0 \\ 0 & 5 & 8 \\ 0 & 0 & 0.2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 6 \\ -3 \\ \frac{39}{5} \end{bmatrix} \quad \begin{matrix} x_1 = 69 \\ x_2 = -63 \\ x_3 = 39 \end{matrix}$$

- Errors & Residuals

Error + Residual

$1.02x_1 + 0.98x_2 = 2$ the true solution is $x_1 = x_2 = 1$

$0.98x_1 + 1.02x_2 = 2$

$$a = \begin{bmatrix} 1.02 & 0.98 \\ 0.98 & 1.02 \end{bmatrix} b = \begin{bmatrix} 2 \\ 2 \end{bmatrix} \text{ vs } \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

let $x_1 = 1.02, x_2 = 1.02$

$$[x_{\text{us}}] = \begin{bmatrix} 1.02 \\ 1.02 \end{bmatrix}$$

$$[e] = [x_{\text{rs}}] - [x_{\text{us}}] = \begin{bmatrix} 1.02 \\ 1.02 \end{bmatrix} - \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 0.02 \\ 0.02 \end{bmatrix}$$

$$[r] = [b] - [a][x_{\text{us}}] = \begin{bmatrix} 2 \\ 2 \end{bmatrix} - \begin{bmatrix} 1.02 & 0.98 \\ 0.98 & 1.02 \end{bmatrix} \begin{bmatrix} 1.02 \\ 1.02 \end{bmatrix} = \begin{bmatrix} -0.04 \\ -0.04 \end{bmatrix}$$

The error and the residual are small

let $x_1 = 2, x_2 = 0$

$$[e] = [x_{\text{rs}}] - [x_{\text{us}}] = \begin{bmatrix} 1 \\ 1 \end{bmatrix} - \begin{bmatrix} 2 \\ 0 \end{bmatrix} = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

$$[r] = [b] - [a][x_{\text{us}}] = \begin{bmatrix} 2 \\ 2 \end{bmatrix} - \begin{bmatrix} 1.02 & 0.98 \\ 0.98 & 1.02 \end{bmatrix} \begin{bmatrix} 2 \\ 0 \end{bmatrix} = \begin{bmatrix} -0.04 \\ 0.04 \end{bmatrix}$$

The error is large but the residual is small

Small residual does not guarantee small error

- Infinity norm & 1-norm

1-Norm

$$\begin{bmatrix} 7 & 1 & 8 \\ 4 & 5 & 8 \\ 10 & 4 & 2 \end{bmatrix} \quad \max \left[(7+4+10), (1+5+4), (8+8+2) \right]$$

$$\max [21, 10, 18]$$

$$\|A\|_1 = 21$$

∞ Norm

$$\begin{bmatrix} 7 & 1 & 8 \\ 4 & 5 & 8 \\ 10 & 4 & 2 \end{bmatrix} \quad \max \left[(7+1+8), (4+5+8), (10+4+2) \right]$$

$$\max [16, 17, 16]$$

$$\|A\|_\infty = 17$$

- 2-norm

2-Norm

Largest eigenvalue of $A^T \times A$ (A^T transpose of A) and calculating its square root

$$A = \begin{bmatrix} 5 & 1 & 1 \\ 1 & 3 & 2 \\ 3 & 4 & 3 \end{bmatrix} \quad A \times A^T = \begin{bmatrix} 35 & 20 & 16 \\ 20 & 26 & 19 \\ 16 & 19 & 14 \end{bmatrix} \rightarrow \text{find eigenvalues } A \times I \lambda = 0$$

$$35 - \lambda, (26 - \lambda)(14 - \lambda) - 361 = 0 = \lambda^2 - 40\lambda + 33 = 0$$

$$(35 - \lambda)(\lambda^2 - 26\lambda - 14\lambda - 3) = 0$$

$$35 - \lambda = 0$$

$$\lambda = 35$$

$$\lambda = \frac{40 \pm \sqrt{1600 - (4 \times 33)}}{2}$$

$$\lambda = -0.07, \lambda = 40.07$$

$$\|A\|_2 = (\lambda)^{\frac{1}{2}} = 6.33$$

- Euclidean norm

Euclidean Norm

$$\begin{bmatrix} 2 & -1 & 5 \\ 0 & 2 & 1 \\ 3 & 1 & 1 \end{bmatrix} \quad \sqrt{2^2 + (-1)^2 + 5^2 + 0^2 + 2^2 + 1^2 + 3^2 + 1^2 + 1^2}$$

$$= \sqrt{45} = 3\sqrt{5} = 6.708$$

- Condition numbers

Condition Numbers

$$\text{Cond}[A] = \|A\| \|A^{-1}\|$$

Cond[A] with Euclidean Norm

$$A = \begin{bmatrix} 3 & 2 \\ -2 & -1 \end{bmatrix} \quad A^{-1} = \begin{bmatrix} -1 & -2 \\ 2 & 3 \end{bmatrix}$$

$$= \sqrt{13} \times \sqrt{17} = \sqrt{221} = 14.869$$